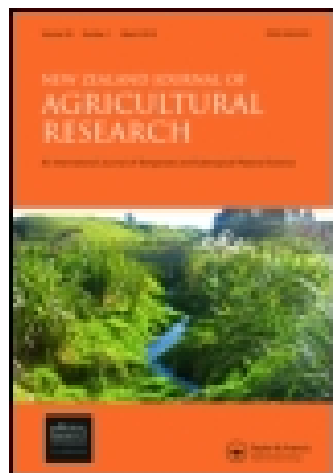


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## New Zealand Journal of Agricultural Research

Publication details, including instructions for authors and subscription information:

<http://www.tandfonline.com/loi/tnza20>

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Published online: 17 Mar 2010.

To cite this article: L. Cransberg & D. J. McFarlane (1994) Can perennial pastures provide the basis for a sustainable farming system in southern Australia?, New Zealand Journal of Agricultural Research, 37:3, 287-294, DOI: [10.1080/00288233.1994.9513067](https://doi.org/10.1080/00288233.1994.9513067)

To link to this article: <http://dx.doi.org/10.1080/00288233.1994.9513067>

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## Can perennial pastures provide the basis for a sustainable farming system in southern Australia?

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**Abstract** Land degradation, associated with farming systems based on annual pastures and crops, is threatening the long-term viability of Australian agriculture. Perennial pastures could play a significant role in reducing some forms of land degradation i.e. dryland salinity, erosion, and soil structure decline. In terms of sustainability, the key differences between perennial and annual pastures relate to rooting depth and the water uptake pattern throughout the year. Deep-rooted perennials can use water when annual pastures are dead, recover leached nitrate (and reduce acidification), and can provide soil cover and root mass to restrict wind and water erosion. Deeply penetrating roots also create biopores and improve soil structure and drainage. In addition, problems normally associated with water repellency are reduced as perennials are not required to germinate each year. There is an urgent need to breed plants that can colonise hostile Australian subsoils characterised by high soil strength, acidity, and high concentrations of toxic elements. In terms of the four perennial grasses considered in this workshop, we believe that the deeper rooting habit of phalaris and tall fescue offer more scope for overcoming degradation than cocksfoot and perennial ryegrass. Plant breeders need to consider the considerable benefits of deep-rooted lines and lines which are able to colonise soils with hostile subsoils.

**Keywords** perennial pasture; sustainable agriculture; plant breeding; production

## INTRODUCTION

In the last decade, the emphasis of agricultural producers in Australia has shifted from a system that only maximises profit to one which includes conservation (Mues & Collins 1993). The change resulted from the realisation that current farming systems are not sustainable, with large tracts of agricultural land becoming unsuitable for traditional agriculture each year. There is an urgent need for widespread land management planning, to ensure the soil resource is not farmed beyond its capability, while meeting economic criteria for survival. In times of depressed commodity prices, and when Australian farmers are forced to assume a price-taking role for their produce, an efficient agricultural system will need to become even more streamlined to ensure long-term survival. It is under these circumstances that our animal production industry will operate. It is likely that temperate perennial grasses, in conjunction with legumes, could provide a sustainable base in southern Australia. It will therefore be essential to ensure that plant breeding and research include sustainability goals. However, it is questionable whether existing selection criteria for temperate grasses address the conservation aspects we need.

A sustainable agricultural system is defined as one which offers continued productivity without degrading either the base resource or ecosystems influenced by the agricultural system.

The main forms of land degradation in Western Australia were identified in House (1991) by the Select Committee on Land Conservation (Table 1). Large areas are already degraded and they are costing the state about \$660 million each year (a conservative estimate as pastoral areas and the cost of eutrophication have not been included). Given the magnitude of this figure, it is clear that developing sustainable agricultural systems is of great national importance.

The aims of this paper are to (i) identify forms of land degradation that may be influenced by perennial pastures, and (ii) specify important

perennial plant characteristics which should be considered in breeding programmes to decrease land degradation in areas where they are planted.

SUSTAINABILITY CONSIDERATIONS

Hydrologic imbalance

In Western Australia, replacing native vegetation with annual crops and pastures has increased percolation of water, caused watertables to rise, and mobilised stored salts resulting in secondary soil salinity (Smith 1962; Peck 1978). The hydrologic imbalance is continuing. For example, the areas affected by secondary salinity in Western Australia increased by 18,000 ha/year between 1979 and 1989 (George 1990). In addition, in 82 bores on the south coast of Western Australia, the water levels rose by an average of 0.43 m/year over 19 years (McFarlane 1992).

Pastures cover most of the agricultural area and therefore even small improvements in their water use could impact significantly on rates of salinisation and the area ultimately affected. There is increasing evidence to suggest that perennial pastures have the capacity to use more water than annual pastures. Measurements of evapotranspiration (Et) and groundwater recharge have shown that annual pastures use less water, resulting in more recharge than annual crops, perennial pastures, or trees (Nulsen & Baxter 1982; Oram & Cooke 1982; Nulsen 1984; Greenwood et al. 1985; Farrington et al. 1992). Carbon et al. (1982) suggested that perennial pastures (*Eragrostis curvula*, *Hyparrhenia hirta*, and *Medicago sativa*) on deep sands were similar to a native forest in

terms of water use patterns, soil moisture depletion over summer and autumn, and deep drainage. In the dehesa system in Spain, Joffre et al. (1988) found that Et for perennial grasses (*Phalaris aquatica* and *Lolium perenne*) was about 200 mm greater than, and recharge only one sixth, that of annual pasture. Scott & Sudmeyer (1993) found phalaris used from 10 to 100 mm more water than annual pasture when grown on a sand-over-clay soil. The phalaris acted as an extended-season annual under these conditions. It is likely that in soils with higher water-holding capabilities, or in years with more summer rain, the difference in water use between annual and perennial pastures will be greater. Clifton (pers. comm.) suggested that phalaris, lucerne, and cocksfoot use similar amounts of water, about 100 mm more per year than annual pastures. Further increases in water use are possible in some species under reduced levels of defoliation during summer and autumn (Taylor & Clifton 1993).

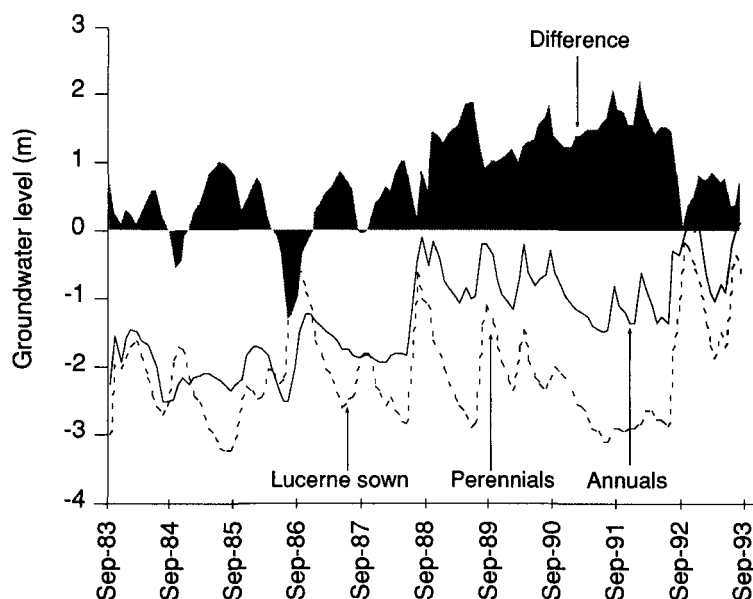
Ferdowsian & Greenham (1992) found that annual pastures and cocksfoot had few roots penetrating deeper than 0.5 m, phalaris and Yorkshire fog roots extended to 1.0 m, and kikuyu roots extended below 1.0 m. Kikuyu had a larger root mass and remained green throughout the summer whereas the other pasture species either died or were unproductive. After considering the amount of water available to plants in several soil profiles, Ferdowsian & Greenham (1992) estimated that annual pastures in an average rainfall year used 58 mm less water than mixed annual/perennial pastures. The perennials used about 60 mm more water than the annuals in December and January but similar amounts during the rest of the year.

Several sets of unpublished data from Western Australia show that perennial pastures use more water than annual pastures. A low-lying area on the south coast in a 380-mm rainfall zone was planted to lucerne (and a 4-row tree belt) in 1987. Groundwater levels under this, and an adjoining paddock under an annual pasture-crop rotation, were similar before the perennials were established (Fig. 1). The difference in groundwater levels increased markedly until two wet winters reduced the difference. A return to more typical rainfalls should re-establish the difference.

In Esperance, Western Australia, soil water profiles over summer and autumn under perennials were compared with those under annual pastures and under cereal and lupin crops (Cransberg unpubl. data). Lupins extracted more water during spring

Table 1 Extent and cost of land degradation problems in Western Australia (House 1991).

| Form of land degradation | Estimated area affected (ha) | Estimated annual cost (\$m/year) |
|--------------------------|------------------------------|----------------------------------|
| Dryland salinity         |                              |                                  |
| 1989                     | 443 090                      | 62                               |
| Potential                | 2 447 000                    | 952                              |
| Waterlogging             |                              |                                  |
| Pasture                  | 1 300 000                    | 90                               |
| Crop                     | 500 000                      |                                  |
| Water erosion            | 750 000                      | 21                               |
| Wind erosion (variable)  | 50 000                       | 21                               |
| Soil structure decline   | 3 500 000                    | 70                               |
| Subsoil compaction       | 8 500 000                    | 153                              |
| Water repellence         | 5 000 000                    | 150                              |
| Soil acidification       | 375 000                      | 5                                |



**Fig. 1** Groundwater levels under lucerne (and some trees) - - - - ; (sown August 1987) relative to annual crops/pastures (—).

and early summer than the perennials, a mixture of cocksfoot, phalaris, tall fescue, and perennial ryegrass in their year of establishment (Fig. 2). Paired (annual and perennial) data are available to only 0.5 m, though the influence of perennials extended to around 1 m. In the top 0.5 m, soil water under annuals increased from 40.7 to 61.1 mm, and under perennials decreased from 54.5 to 34 mm between December 1992 and April 1993. Rainfall during this period was 143 mm. The comparison between an annual pasture and an established veldt grass pasture showed smaller differences, probably because veldt grass has shallower roots. Total soil water storages were lower under the perennials at all times with differences in early January, late March, and late April, of 22, 46, and 4 mm water/m soil, respectively.

Perennials are able to dry soil profiles to a greater depth than annual pastures, resulting in less waterlogging during the subsequent winter. Johnston (pers. comm.) reported that the pre-winter water deficit to 1 m depth under lovegrass was satisfied (attaining field capacity) 69, 62, 110, and 80 days later than under an annual pasture, from 1990 to 1993 respectively.

### Soil acidification

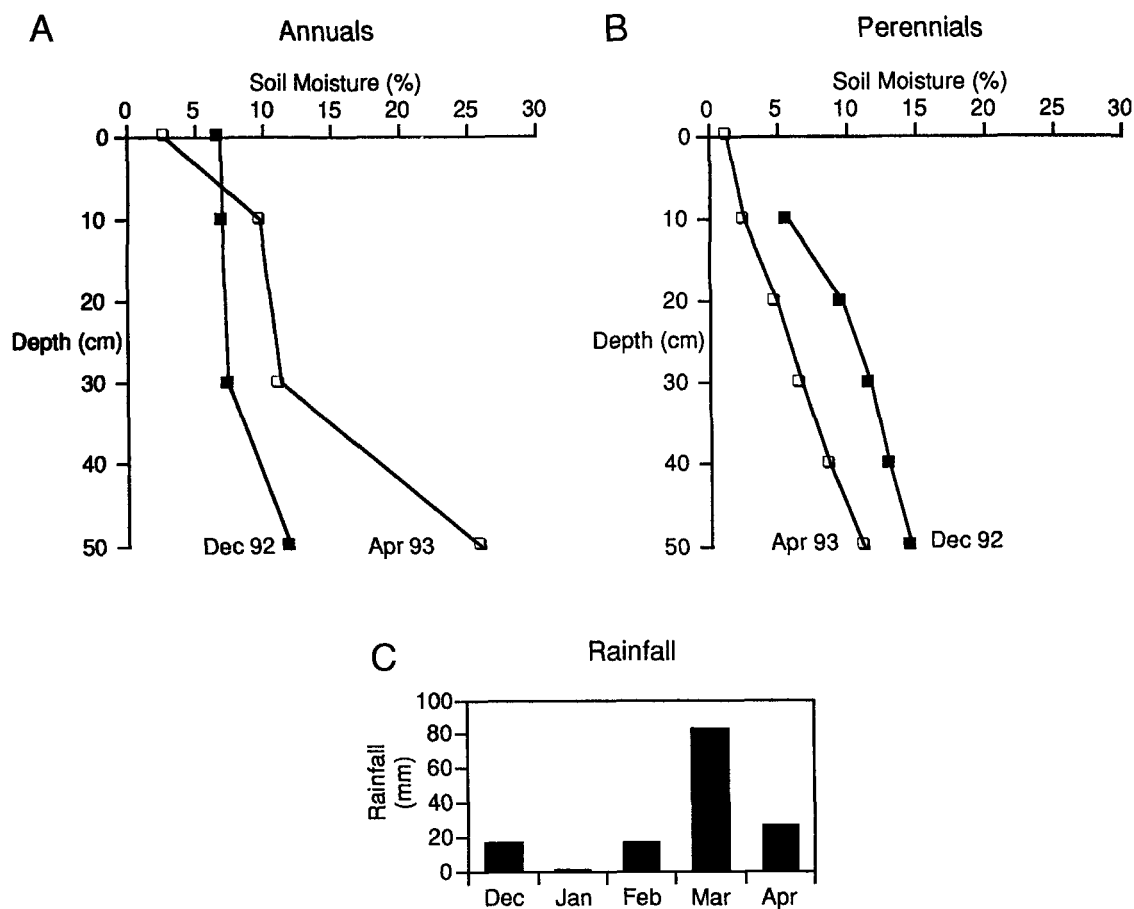
Increasing soil acidity has been linked with declining productivity over 14 million hectares of southern Australia (Coventry 1985). Products from

the biological carbon and nitrogen cycles cause the acidification (Helyar 1976). Perennial pastures have the potential to influence rates of acidification. Nitrate accumulates over summer when the subterranean clover plants break down. In annual pastures, the nitrate leaches from the profile following the opening autumn rains before the annual pasture can develop an actively growing root system (Simpson 1962). Summer-active, deep-rooted pasture species can reduce the accumulation and leaching of nitrate (Ridley et al. 1990; Oram et al. 1993). Work conducted in north-eastern Victoria showed nitrate leaching still occurred under phalaris pastures, though the level was lower than under annual pasture, as was the rate of acidification to 0.6 m (Ridley et al. 1990). The capabilities of other perennials in reducing accumulation and leaching of nitrates need to be assessed.

While acidification rates may be reduced under deep-rooted species, the selection of plant species capable of tolerating low soil pH and high levels of soil aluminium will continue to be important. Perennials may extend the time before lime needs to be added and increase the prospects of success in plant breeding and selection programmes for acid-tolerant cultivars.

### Wind and water erosion

Plant cover is probably the most important factor affecting how much soil is lost by both water and



**Fig. 2** Soil water changes under (A) an annual lupin crop and (B) a perennial grass pasture, and on-site rainfall (C), between December 1992 and April 1993.

wind erosion. For wind erosion, standing and prostrate cover standards have been established for some soil types (Findlater et al. 1990; Leys & Heinjus 1991). Much of this work has been conducted following cropping, where soils are exposed after harvest. Subterranean clover and capeweed components of annual pastures are readily fractured in grazed situations and plant residue levels are usually low by autumn. Annual grass residues, which are usually prostrate, are not broken down as readily but decline over summer through animal and microbial consumption.

Standing plants increase the roughness of the surface exposed to wind, thereby reducing the erosivity of the wind (Chepil & Woodruff 1963). A pasture based on perennial grasses should provide

both increased roughness (to prevent high-velocity winds reaching the soil surface) and a soil-binding effect with its root system.

Plant cover is also critical for controlling water erosion. Plants protect the soil from detachment by raindrops and reduce the velocity of runoff. Once ground cover falls below about 75%, discrete bare areas begin to interconnect and runoff rates rise dramatically (Lang 1979; Lang & McCaffrey 1984).

### Water repellence

Water repellence (caused by the build-up of repellent organic matter in sandy soils), slows the penetration of water into dry soils, and is increasing under improved pastures and cropping systems in temperate Australia (King 1981). Germinating

pastures are patchy and there is little soil moisture for shallow-rooted plants (Bond 1972). Pasture growth is delayed, soils are vulnerable to wind and water erosion for a greater period, and seeds which may germinate on the surface extend their roots into dry soil and die. This is particularly true for plants with large seeds (e.g., subterranean clover) whereas lighter, more mobile seeds (of plants such as capeweed and silver grass) accumulate in low points where there is a greater chance of survival. In time these undesirable species dominate the pasture (King 1985). The role of perennials is limited in regard to combating water repellence. Perennial plants, once established, have the advantage of not having to germinate each year and can reduce wind erosion and restrict weed invasion. However, water repellency has been reported to increase under long-term lucerne and veldt grass stands (King 1985).

### Eutrophication of waterways

Sediments and nutrients lost from annual pastures are contributing to the silting and eutrophication of waterways. Nutrients are usually topdressed in autumn, before the pasture has developed an active root system. There is usually little plant cover at this time and runoff associated with the opening rains carries fertilisers and topsoil enriched in nutrients. During one autumn storm near Albany, up to 60% of applied phosphorus (P) was lost in particulate form (Weaver & Prout 1993). Knight (1990) reviewed the role of perennial plants in reducing the movement of P from pastures to waterways. He suggested that perennials act as a vegetative barrier, filtering sediment and particulate P and reducing overland flow. Further, the deep, established root system can take up labile P in autumn, thereby reducing P losses to waterways. Other nutrients leached beyond the shallow root zone of annual pastures can be used and recycled by deep-rooted perennial plants.

### Soil structure decline

Many Australian subsoils are sodic resulting in high soil strengths, poor drainage, and poor aeration (Rengasamy & Olsson 1991). These authors showed that in sodic soils it is necessary to replace the sodium with calcium before increases in organic matter can improve soil structure. The ability of plants to create biopores in hard soils depends on the ability of the roots to thicken in response to soil strength, thereby increasing the axial pressure applied (Materechera et al. 1991). The wide

variability among species to penetrate soils is encouraging for plant breeders.

Roots from lucerne penetrated the domes in a solanised solonetz soil, resulting in less water-logging and better growth of annual pasture species and crops in following years (Hurst & Hurst 1992).

## PRODUCTIVITY CONSIDERATIONS

### Sheep production

Animal requirements also need to be closely considered in sustainable agricultural systems. The provision of green feed by perennials over summer and autumn is extremely valuable for grazing animals (Orsini 1989), but defoliation also decreases transpiration and may reduce plant persistence. Management systems that maximise total pasture production throughout the year will result in the highest water use. This does not necessarily mean low stocking rates. Korte (1987) suggested that long periods of extremes of pasture mass should be avoided to maintain both pasture quality and productivity. The two critical periods are during spring when biomass accumulation is high, and after dry periods in summer and autumn when pastures can be overgrazed. Management should be flexible to cope with seasonal differences, to achieve agronomic goals for particular pasture species, and to achieve animal production goals. The management to achieve agronomic goals will not always be the same as that needed for animal production goals and compromises will be needed.

## FACTORS FOR SCREENING IN BREEDING PROGRAMMES

The key factors governing plant water use potential that require screening in plant breeding programmes are effective rooting depth, leaf area index (LAI), and summer activity given sufficient soil moisture. Rooting depth has been correlated ( $r^2 = 0.75$ ) with soil moisture depletion (Brown et al. 1983) and with Et ( $r^2 = 0.31$ ; Scott & Sudmeyer 1993). Selecting plants to maximise water use will entail either a deep root system to allow transpiration to continue through the dry summer, or a capacity to produce biomass from stored soil water or in response to "out of season" rainfall (when annual pastures are dead). Plants with a C4 pathway should also be considered, being summer-active and having higher water use efficiencies at high temperatures.

They also have better drought stress capabilities (Levitt 1972).

While the cost may prohibit screening for rooting depth in plant breeding programmes, species evaluation should be conducted on soils which are known recharge areas (or are aimed at other forms of degradation). If they grow well in these soils then they will be using water that may add to saline groundwaters.

## DISCUSSION

Developing sustainable agricultural systems should be done in a prioritised and structured manner and implementing these systems will be easier if there are no radical changes needed. Of the sustainability issues reviewed in this paper, the highest priority is water use. Without a system that uses more water, rising water tables and salinisation will continue. Our current range of plant species for salt-affected, waterlogged land is extremely limited and unlikely to be economic given current commodity markets. Acidification is the second priority, with a reasonable range of agronomic options for intermediate levels of soil acidity and the option of liming to reverse the problem. Subsoil acidification is a more major concern as it is much more expensive to rectify.

Plant breeding targets have been traditionally linked to production goals, for example disease and insect resistance, improved seedling vigour, increased dry matter yield, growth habit and seasonal productivity patterns, better drought tolerance, reduction in plant toxins harmful to livestock, and physical characteristics of seed and seedheads to minimise wool contamination. These are necessary targets, capable of demonstrating large dollar benefits on a national basis. The environmental aspects of our pasture production systems have received much less consideration. One recent exception has been the breeding of acid-tolerant phalaris lines (Oram et al. 1993). It is now imperative for plant breeders to include the sustainability aspects of pasture plants in their breeding programmes to address land degradation problems.

The main constraints to the adoption of perennial pastures are the perceived cost and uncertainty of establishment, and their poor persistence under heavy or continuous grazing regimes (especially sheep grazing). Farmers need to modify their grazing management and there is a reluctance to do this. The high likelihood that they

will lose substantial parts of their farm to salinity is an important consideration for farmers adopting perennial pastures in regions of Western Australia.

Trees by themselves are unlikely to prevent salinity in many parts of southern Australia as the drawdown around trees is minimal in heavy clay subsoils, particularly when the groundwater is saline. In addition, recharge is thought to be important over most of the landscape and it is not possible to prevent salinisation by planting small high intake areas, or to intercept all of the laterally flowing groundwater as it is often too saline or too deep. Therefore there is a need for broad-scale treatments such as perennial pastures *in addition to* strategically planted tree belts.

Funding environmental research is expensive because it is long-term and requires a multi-disciplinary approach. This research is almost entirely funded by government bodies and coordinating them will be necessary to allow the multi-disciplinary approach to meet sustainable goals. There is also scope to incorporate industry funds into the development of sustainable production systems. First, some measures of sustainability will need to be developed for objective evaluation purposes. Sustainability is best measured on a large scale, either a catchment or other relevant area, and should measure changes in factors we perceive to be important for a sustainable system e.g., areas of vegetation, nutrient loads in major waterways and land affected by salinity.

## ACKNOWLEDGMENTS

Helpful comments were received on this paper by Dr Bob Nulsen and Dr Jo McFarlane.

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